

Range Queries and Copies

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to maintain a list of arrays which initially has a single array. You have to process the following types of queries:

1. Set the value a in array k to x .
2. Calculate the sum of values in range $[a, b]$ in array k .
3. Create a copy of array k and add it to the end of the list.

Input

The first input line has two integers n and q : the array size and the number of queries.

The next line has n integers $t_1, t_2, \dots, t_n$: the initial contents of the array.

Finally, there are q lines describing the queries. The format of each line is one of the following: "1 k a x ", "2 k a b " or "3 k ".

Output

Print the answer to each sum query.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq t_i, x \leq 10^9$
- $1 \leq a \leq b \leq n$

Example

Input	Output
5 6 2 3 1 2 5 3 1 2 1 1 5 2 2 1 5 1 2 2 5 2 1 1 5 2 2 1 5	13 13 13 15